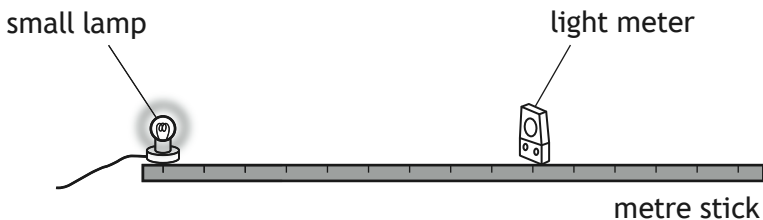


13. A small lamp is placed 0.40 m from a light meter in a darkened room.



The irradiance of the light from the lamp is measured using the light meter.

The irradiance at this distance is  $20 \text{ mW m}^{-2}$ .

The light meter is now moved until the irradiance is  $80 \text{ mW m}^{-2}$ .

The distance between the lamp and the light meter is now

- A 0.04 m
- B 0.10 m
- C 0.20 m
- D 0.80 m
- E 1.60 m.